



a racer in different levels. Game Maker uses two basic components for making games. The first type is static, and the second type is dynamic. Static components are sprites, sounds and backgrounds. A sprite can be any element in the game, the ground, the trees, the enemies and the player are all defined using sprites. Dynamic components include objects and rooms. Objects are sprites with attributes and behaviour.

An orange orb used as a bullet is a sprite, but an orange orb programmed to come into existence when the [Space] key is pressed, produces a sound for that action and kills enemies, is an object. A room is the game environment using the background and any object that the user puts in.

Discussed below is a method to create a simple top-down shooter. The logic and approach to handling Game Maker can be used for creating any type of game.

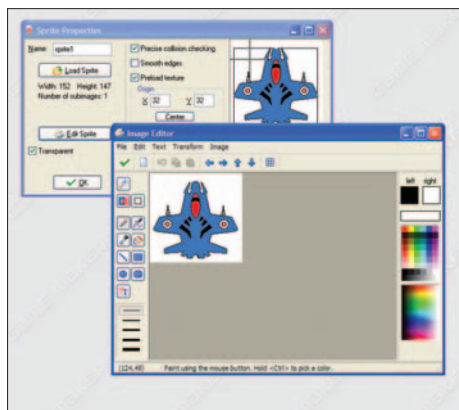
The first step is to create and load all the static objects. This means creating sprites, backgrounds and sounds. Sprites can be created in paint or any image creation program of your choice. Use very small images, and zoom in and work. Simple lines are enough to create sprites. We used paint to create this space ship.

Think of all the elements you'll need in the game. We created enemy ships, and bullets. We also used a stock explosion animation.

Animated GIF files can be used in Game Maker, and they will be animated in-game as well. We also used sounds downloaded from the internet, and a space wallpaper for the background. Reduce the quality of large backgrounds. There are a wealth of sprites, sounds and backgrounds available for free, for any kind of game that you may want to create. Sprites are

often drawn with different views in the same file, so you will have to break them down into individual images if you are not using your own.

Fire up Game Maker 7 now. You will see a number of folders on the left hand side. We start by loading our sprites. Right-click on Sprite, and then click on Create Sprite. A window should appear. Click on load, navigate to the image file of your sprite and click on Open. This window also has the sprite editor, which lets you change colours, and resize the sprite. Typically, make a sprite smaller than 100 pixels.



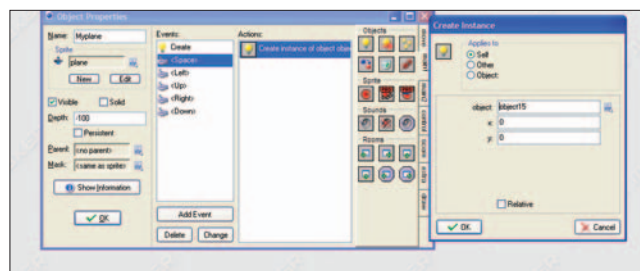
Edit sprites

Your sprites will be named sprite, sprite1, sprite2, and so on. Change these to more recognisable names that are simpler to handle if you have a lot of sprites. Load all the sounds you want to use too. You can have sounds for every bullet fired, explosions, and also have a background score too. You can also have sounds for the movement of your space-

ship towards the left or right. All this will be programmed later on. Next, load the background.

The right programming makes all the difference in the game. Game Maker can ascribe behaviour to the object based on the actions of the user. To create a new object, right-click on the Objects folder. Then, load a sprite in the Sprite window by clicking on an option from the drop-down box. Once you have made an object, changes to the sprite will change the object automatically. Now create a new event.

On the right-hand side pane, in a tabbed view are a number of options. Drag and drop these actions as per your requirements. You can play a sound, make the object move, jump to a particular position, change into another image and even make actions based on probability. Explore these actions - you don't need all of them. Add events for all the movement keys, and move the object to the appropriate directions. Now you will have to program the fire key. For this, you will have to create another object, which is the firing object. Make the object move towards the top of the screen on creation. Go back to the plane object, and add an event for the [Space] or the [Ctrl] key. In the main tab is a create instance button (a lightbulb). Click on this and select the instance of the fire object. Now, every time you press [Space], your space ship will spew fire.

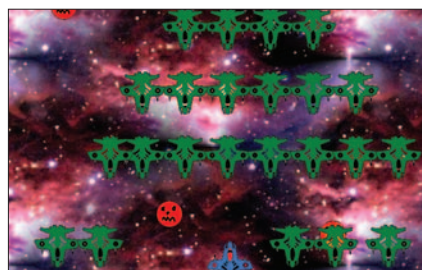


Create instances

Create enemy objects, and configure them to vanish when they get hit by the bullets that are fired. You also have to make sure that your own plane gets destroyed when it is hit.

Now create a new room. Load the background for the room in the backgrounds tab. The drop down box will show all the loaded backgrounds. Check the horizontal and vertical tile options. Click on Views, and check Enable view. Now, select the dimensions of the game. This can be 400 x 300. You can make exotic scrollers here, such as a 200 x 200 square scroller. In the View in Room box, set the Y value to a large

number (more than 2000). This lets you place enemies to appear when playing the game. Place a few enemies, and you're good to start playing. However, you need a lot of tweaking before you can perfect a game. The real building starts from here.



Finally the game

Be dedicated, and have fun for fruitful results. Game Maker has some scripting capabilities too. Once you've understood the basics of how gameplay works, you can move on to better things. 3DRad, is another software that lets you create 3D games in the same spirit as Game Maker. ■

aditya.madanapalle@thinkdigit.com